Bachelor of Technology
Computer Science and Engineering

Submitted By

RAGYAM SINHA
13000123064

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Techno Main
EM-4/1, Sector-V, Salt
Lake Kolkata- 700091
West Bengal
India

Assignment Report: -
Location Transparency in Social Media

1 Introduction

In simple words, the question is asking: "How do giant apps like Facebook or Instagram hide the fact that their data is spread across thousands of servers all over the world?" When you look at a friend's photo, you don't have to know if that photo is stored in a server in USA, Singapore, or Mumbai. You just click "view." Location Transparency is the magic technology that makes the whole world feel like one single database.

2 Keywords

To understand this topic, we need to look at these 4 keywords:

- Data Directory (Catalog)
- Naming Transparency
- Replication
- Distributed Query Processing

3 Elaboration on Keywords

3.1 Data Directory (Catalog)

In a DDBMS, a directory is like a phonebook. It keeps a record of which data "piece" is sitting on which "machine." For social media, the directory maps your User ID to a specific data center.

3.2 Naming Transparency

This means that the name of a file or a post does not change based on where it is stored. Your profile picture has one unique ID, and it stays the same whether the system moves it to a server in Europe or Asia.

3.3 Replication

Social media platforms copy your data (like your posts) to many different countries. This is done so that if you travel to London, the app can show you your data from a nearby server instead of pulling it from India.

3.4 Distributed Query Processing

This is the brain of the system. When you search for a friend, this system breaks your search into small tasks and sends them to the right servers without you ever knowing.

4 Utility in Social Media

Why do we need these? 1. **User Experience:** If users had to type IP addresses of servers to see a photo, no one would use the app. 2. **Speed:** By hiding locations, the system can automatically pick the closest server to the user. 3. **Reliability:** If the USA server goes down, the system uses the location transparency map to find the same data in a backup server.

5 Final Explanation: How it works in DDBMS

Global social media platforms use a “Multi-level Schema Architecture.”

1. **The Global Schema:** This is what we see. It looks like one big table of “Users” and “Posts.”

2. **The Mapping Layer:** This is where Location Transparency lives. When a query comes in, the DDBMS looks at its **Global Conceptual Schema**. It sees that the user wants “Post A.”

3. **Physical Layer:** The system then looks at the **Local Internal Schema** to see that “Post A” is physically sitting in Server #509 in North Carolina. The system “resolves” the name to a physical address in the background. The user only sees the “Logical Name” of the post.

6 Real-World Examples

- **Facebook (Cassandra/TAO):** Facebook uses a system called TAO. When you check your notifications, you don’t know which data center is answering. The TAO layer handles the location for you.
- **YouTube:** Videos are stored in “Edge Servers” (Content Delivery Networks). When you click a video, the URL looks the same, but the system pulls the data from the server closest to your city.
- **WhatsApp:** Your messages are stored in a distributed way, but your chat list looks like it’s all in one place.

7 Conclusive Summary

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Location Transparency is the reason why global social media feels so fast and simple. In a DDBMS environment, it is ensured by separating the "Logical Name" of the data from its "Physical Address." By using directories and smart query routing, platforms like Instagram hide the messy reality of global servers, allowing users to focus only on their content.

9 Academic References

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2. **Coulouris, G., Dollimore, J., & Kindberg, T. (2011).** *Distributed Systems: Concepts and Design*. Pearson Education.
3. **Silberschatz, A., Korth, H. F., & Sudarshan, S. (2020).** *Database System Concepts*. McGraw-Hill.

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